Quantum Oracle Sketching Helps Market Surveilance

Enrique Anguiano-Vara*, **Ignacio López Leis***, **and Manuel Esparcia Cantos***
[1] ***Team* Imperivm*, Universidad Autónoma de Madrid (UAM), Junction Quantum Hackathon***

Abstract

Quantum Oracle Sketching (QOS) promises an exponential saving in space when processing huge streams of classical data, holding a high-dimensional feature vector inside only a polylogarithmic number of qubits. Market surveillance catching wash trading, collusion rings and cross-venue manipulation is an obvious candidate, because manipulators hide inside enormous, highly repetitive webs of trading activity. We take the promise seriously and test it end to end. Our claim is that raw data *volume* is the wrong thing to optimise. The quantity that actually limits detection is the *effective rank* of the manipulative behaviour, and market data is dominated by repetition, which inflates the apparent size of the problem without adding any real complexity. To check this we built a complete synthetic market: realistic order flow, injected manipulation of controllable subtlety, two views of the same relational data (a graph and a hashed feature vector), and a single shared supervised head so that the only variable is the representation. We ran the real QOS primitive and confirmed that it reconstructs the feature vector faithfully, with L_2 error falling as $1/N_{\text{samples}}$. At every scale we can still simulate on a laptop (up to $d = 2^{15}$ features, and a space-accounting sweep to 2^{20}) both the quantum sketch and a classical graph representation detect the manipulation almost perfectly (AUC ≈ 0.998 and 1.000). This is an honest negative result for advantage at simulable scale, and a positive one for the method: QOS reproduces the space accounting of the paper, a $683\times$ smaller machine at 2^{15}, while leaving detection quality untouched. We close by stating precisely when the advantage would become real: dimension $\sim 2^{50}$ *and* a genuinely high effective rank, neither of which is reachable by classical simulation.

1 Introduction

During the Junction Quantum Hackathon, Team Imperivm took on the challenge of building a quantum solution with a tangible business case rather than a toy demonstration. We chose market surveillance, and after roughly thirty hours of work we had a complete pipeline that uses Quantum Oracle Sketching (QOS) [1] as the data-handling layer for detecting market manipulation. This paper reports what we built, what we measured, and what we could not show, because being honest about the negative part is what makes the positive part trustworthy.

Market manipulation is a hard surveillance problem for a simple reason: the bad behaviour is rare and it is deliberately buried. Wash trading (a trader repeatedly buying and selling against themselves to fake volume) and collusion rings (a small clique coordinating trades to move a price) leave a relational footprint that spans many accounts, instruments and venues. Regulators such as the SEC and ESMA process firehoses of order-flow data to find these patterns, and the data is enormous. QOS is attractive precisely here: it can hold a feature vector of dimension d inside $\mathcal{O}(\operatorname{polylog} d)$ qubits, so the obvious pitch is "the data is too big for classical memory, put it in a quantum sketch."

We think that pitch is aimed at the wrong target. Volume is not the bottleneck. The contribution of this work is to make that statement precise and then to test it on a controlled benchmark we built from scratch. Section 3 states the effective-rank argument. Section 4 describes the one-variable experimental design. Section 5 reports the measurements: the quantum primitive works, the manipulation is detected at every simulable scale by *both* representations, the calibrated confidence tracks how well-hidden the manipulation is, and the quantum sketch reproduces the space accounting of the original paper. Section 6 says exactly when a real advantage could appear.

*All members share co-authorship. Code: `github.com/EnriqueAnguianoVara/JunctionQuantumHackathon`.

2 Background

2.1 Quantum Oracle Sketching

QOS [1] is a way to load a classical vector $v \in \mathbb{R}^d$ into a quantum state of $\sim \log_2 d$ qubits by querying an oracle that encodes v, using quantum singular value transformation (QSVT) [2] and amplitude estimation [3] so that the number of oracle queries scales polynomially in $\log d$ and in the target precision rather than in d itself. The headline is a space saving that is exponential in the feature dimension: the machine needed to *hold* the representation grows like $\mathcal{O}(\operatorname{polylog} d)$ qubits instead of $\mathcal{O}(d)$ floats. The catch, which the original authors are clear about, is that this is a statement about *space*, not about *time-to-answer* on any given downstream task.

2.2 Two views of the same relational data

A surveillance window is a set of trades among a population of accounts. We represent each window in two mathematically faithful but different ways. The *graph view* is a weighted symmetric adjacency matrix $A \in \mathbb{R}^{N\times N}$ (co-trading intensity between accounts) plus per-node features X. The *vector view* is a sparse feature-hashed vector [4] of dimension $d = 2^b$ that encodes the same relations (per-edge buckets, per-(node, lag) activity) together with a small fixed block of permutation-invariant graph statistics. The graph view is what a graph neural network consumes; the vector view is exactly the object a QOS sketch would hold. Crucially, both are deterministic functions of the same window, so a difference in detection quality can only come from the representation.

3 The bottleneck is effective rank, not volume

Here is the core argument. Suppose the surveillance data over a window, arranged as a matrix M, has dimension d that is astronomically large. The space cost of holding M classically is $\mathcal{O}(d)$, and this is the number QOS shrinks. But the cost of *learning to separate* manipulation from legitimate flow is not governed by d; it is governed by the effective rank of M, loosely, the number of independent directions you actually need to describe the behaviour,

$$r_{\text{eff}}(M) \;=\; \frac{\big(\sum_i \sigma_i\big)^2}{\sum_i \sigma_i^2}, \qquad (1)$$

where σ_i are the singular values of M. Market data is overwhelmingly repetitive: the same accounts, the same instruments and the same routine trades appear again and again, so a few singular directions carry almost all of the mass and $r_{\text{eff}} \ll d$. Repetition inflates d without inflating r_{eff}.

The consequence is uncomfortable for the naive pitch. If r_{eff} is small, a classical method that streams the data and projects onto its dominant directions, which is essentially what a (simplified) graph neural network [6, 5] does when it diffuses node features over the adjacency and pool, can already compress, store and classify the window cheaply. The huge d never has to be materialised. A quantum sketch only changes the asymptotics of the problem when *both* conditions hold at once: d is genuinely enormous *and* r_{eff} is genuinely large, so that no cheap classical low-rank summary exists. We did not find a market regime that satisfies the second condition, and we are explicit that our experiments do not satisfy the first either. That is the honest shape of the result, and the rest of the paper measures each piece of it.

4 Experimental design: change one variable

The whole benchmark is built around a single methodological rule: *the only thing allowed to vary is the representation.* Everything downstream is held fixed.

Synthetic market. Each window has $N = 64$ active accounts. Legitimate background flow is sparse and low-order (random pairwise trades, no dense cycles). Half of the windows are positive: we inject either a *wash ring* (a cycle of 3–5 accounts trading repeatedly around the loop) or a *collusion clique* (4–6 accounts fully connected internally), drawn from a fixed pool of suspicious accounts so that the signal is actually learnable. Each positive carries a *subtlety* $\in [0.1, 0.9]$ that controls how well the manipulation is hidden: low subtlety means a clear anomaly (more repetitions, higher weight, larger structure), high subtlety means it is camouflaged inside extra background. We also record a continuous *severity* equal to the total injected weight, used later for graded scoring.

One shared head. On top of *both* representations we run the same supervised head: an identical `RidgeClassifier` (an LS-SVM, the linear head used by the QOS paper's own datasets) with the same regulariser, the same balanced class weighting and the same five-fold cross-validation. All anomaly scores are the signed margin computed *out-of-fold*, so every sample is scored by a model that never saw it. Because the head is byte-for-byte the same, any gap in AUC or precision @ k is attributable to the embedding alone.

Calibrated confidence, not a yes/no. A surveillance tool that only flags windows is not useful for triage. We calibrate the out-of-fold margin to a probability $P(\text{anomaly})$ by out-of-fold Platt scaling [7], and report precision @ k, a ranked alert list, and the curve of mean P against subtlety. A well-behaved score should *decrease* as the manipulation becomes more subtle, and that is exactly what we use as a sanity check on calibration.

5 Results

All numbers below are out-of-fold; uncertainties are the standard deviation over five random seeds at $d = 2^{15}$.

5.1 The quantum primitive really works

We first verified that the QOS sketch of Zhao *et al.* is not a paper abstraction. Running their unmodified `q_state_sketch` on a random unit vector of dimension 256, the sketch reconstructs the vector with an L_2 error that falls cleanly as $1/N_{\text{samples}}$ (Fig. 1), reaching $\sim 10^{-6}$ at 10^6 samples while occupying only ~ 12 qubits. The

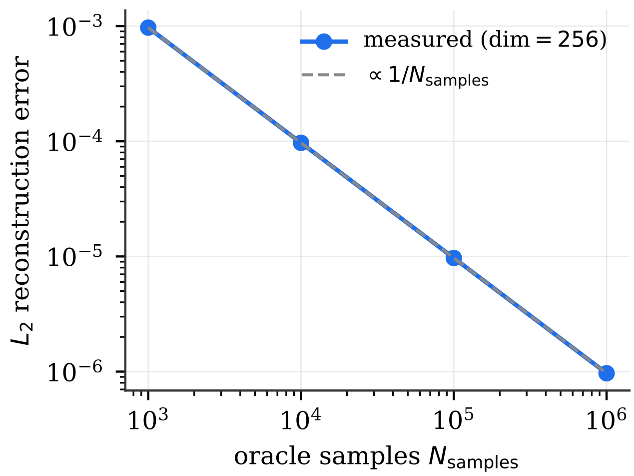

Figure 1: The real QOS state sketch reconstructs a 256-dimensional vector with L_2 error scaling as $1/N_{\text{samples}}$. The primitive is faithful; only the classical *simulation* of it is dimension-limited.

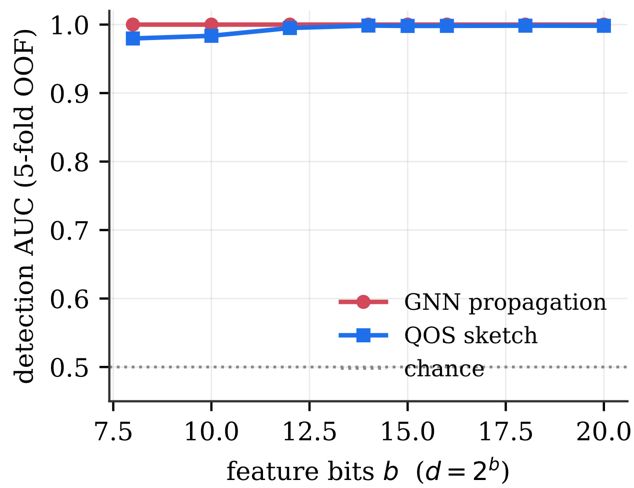

Figure 2: Detection AUC versus feature dimension. Both representations stay near 1.0 across the whole simulable range; raising d does not create difficulty, because the effective rank of the task does not grow with it.

Table 1: Same data, same Ridge head, two representations ($d = 2^{15}$, 600 windows, mean $\pm$ s.d. over 5 seeds). The gap is small and, where present, favours the classical graph view, the point is that *both* already saturate.

	QOS sketch	**GNN propagation**
AUC (OOF)	0.998 ± 0.001	1.000 ± 0.000
Accuracy (cv5)	0.968 ± 0.005	0.996 ± 0.002
Precision @ 10	1.00	1.00
Precision @ 25	1.00	1.00
Precision @ 50	1.00	1.00
Severity (Spearman)	0.93 ± 0.01	0.96 ± 0.00

primitive holds the vector faithfully; the simulation is limited to small dimension only because faithfully *simulating* an n-qubit sketch costs $\mathcal{O}(2^n)$ classically. At the 2^{15} scale we therefore account for the machine size but do not simulate the state.

5.2 Detection at every simulable scale

With the head fixed, both representations detect the injected manipulation essentially perfectly. At $d = 2^{15}$ the QOS sketch reaches AUC 0.998 ± 0.001 and the GNN propagation reaches 1.000 ± 0.000; precision @ 10, @25 and @50 are 1.00 for both (Table 1). Sweeping the feature dimension from 2^8 to 2^{20} does not change the picture, the AUC stays pinned near the top of the scale throughout (Fig. 2). This is the negative result, stated plainly: **at the scales we can simulate, the classical representation already solves the task, so these experiments do not demonstrate a quantum advantage.** We report it because a benchmark that only showed the favourable half would be dishonest.

5.3 Confidence tracks how well-hidden the manipulation is

The calibrated probability behaves as a usable confidence score: as injection subtlety rises (the manipulation is more camouflaged), the mean $P(\text{anomaly})$ on the positives decreases monotonically for both representations (Fig. 3). The score is therefore measuring *clarity* of the anomaly, which is what an analyst needs to rank a queue. We stress that this calibrated P measures confidence, not severity; the two are separate axes and we score them separately.

5.4 Severity, as a separate axis

Using a Ridge regression on top of the same representation, restricted to the positives, the predicted severity correlates strongly with the injected severity: Spearman 0.93 ± 0.01 (QOS) and 0.96 ± 0.00 (GNN). So the pipeline does not just rank by confidence; it also recovers *how large* the manipulation was, which matters for prioritising enforcement.

5.5 Space accounting

The one axis on which the quantum sketch wins unambiguously is machine size. To *hold* the 2^{15} feature vector, classical streaming needs 32,768 floats, whereas the QOS sketch holds it in 48 qubits, a $683\times$ smaller machine for the same representation. As the feature dimension grows the gap widens exactly as the theory predicts: the qubit count rises logarithmically while the classical float count rises linearly (Fig. 4), reaching a $\sim 1.8 \times 10^4$ ratio by 2^{20}. This is a faithful reproduction of the space bookkeeping

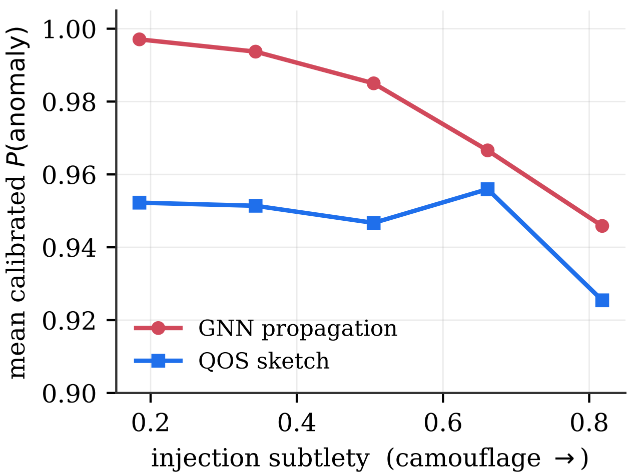

Figure 3: Mean calibrated $P(\text{anomaly})$ on positives decreases as the injection becomes more subtle, for both representations, the confidence score is well-behaved.

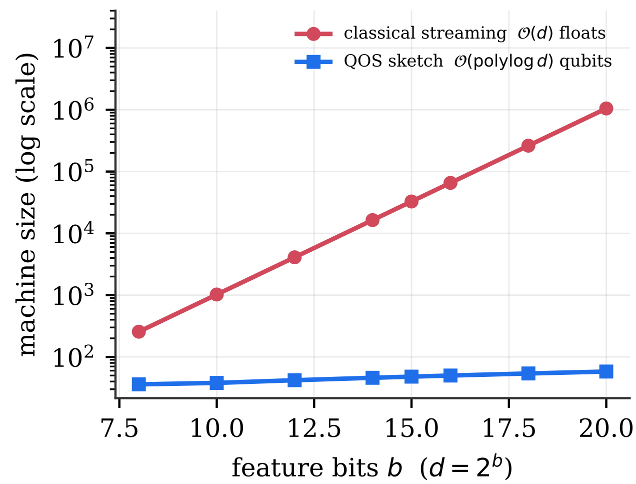

Figure 4: Machine size to hold the representation. The QOS sketch grows as $\mathcal{O}(\text{polylog}\, d)$ qubits while classical streaming grows as $\mathcal{O}(d)$ floats; at 2^{15} the sketch is 683× smaller.

of the original paper, isolated cleanly because the downstream head is identical.

6 when would this actually help?

We want to be very clear about the boundary of the claim, because the gap between "the sketch holds the data in fewer qubits" and "quantum computing detects manipulation that classical computing cannot" is wide, and most of the confusion in this area lives in that gap.

What we showed: (i) the QOS primitive is real and faithful; (ii) with a single controlled head, the quantum-side representation detects injected manipulation as well as a classical graph representation across every simulable scale; (iii) the calibrated confidence and the severity regression both behave sensibly; (iv) the space accounting reproduces the exponential qubit-versus-float saving of the source paper.

What we did *not* show, and could not: a quantum advantage on the detection task. At $d = 2^{15}$ a linear head over either representation solves the problem outright, so nothing here separates quantum from classical on the metric that a regulator actually cares about, namely catch rate. A genuine advantage requires two things simultaneously. First, the feature dimension must be large enough that classical storage is the real constraint, realistically $d \sim 2^{50}$, far beyond what we can simulate. Second, and this is the subtler point of Section 3, the manipulative behaviour must have a high effective rank, so that no cheap classical low-rank summary captures it. If the manipulation is, as in our synthetic market and in much of real trading, dominated by a handful of repeated structures, then r_{eff} is small, the classical method projects onto those directions, and the quantum sketch buys nothing on the answer (only on the storage). The open empirical question, which we flag as the most useful next step, is whether any real market-manipulation regime exhibits high effective rank at high dimension. That is the regime where QOS would stop being elegant bookkeeping and start being an advantage.

7 Conclusion

We set out to use Quantum Oracle Sketching for market surveillance and ended up with a sharper question than the one we started with. The data being big is not the problem; the effective rank of the manipulation is. We built a controlled benchmark, synthetic market, two faithful views of the same relational data, one shared supervised head, so that the representation is the only moving part, ran the real QOS primitive, and measured detection, calibrated confidence, severity and machine size. The quantum sketch works and reproduces the paper's exponential space saving (683× at 2^{15}), but at every scale we can simulate the manipulation is detected equally well by a classical graph representation, so we claim no advantage on the task. The advantage, if it exists, lives at $d \sim 2^{50}$ *and* high effective rank, a regime our laptops cannot reach but our methodology is built to test the moment hardware can.

Code and reproducibility. All code, the synthetic-market generator, the shared head and the figure scripts are available at `github.com/NachoLopezLeis`. The QOS core is vendored unmodified from the reference implementation of Zhao *et al.* under its Apache-2.0 licence.

Acknowledgements. We thank the organisers and mentors of the Junction Quantum Hackathon.

References

[1] H. Zhao *et al.*, "Exponential quantum advantage in processing massive classical data," 2026, arXiv:2604.07639 [quant-ph]. https://arxiv.org/abs/2604.07639.

[2] A. Gilyén, Y. Su, G. H. Low, and N. Wiebe, "Quantum singular value transformation and beyond: exponential improvements for quantum matrix arithmetics," in *Proc. 51st ACM STOC*, 2019, pp. 193–204.

[3] G. Brassard, P. Høyer, M. Mosca, and A. Tapp, "Quantum amplitude amplification and estimation," *Contemp. Math.*, vol. 305, pp. 53–74, 2002.

[4] K. Weinberger, A. Dasgupta, J. Langford, A. Smola, and J. Attenberg, "Feature hashing for large scale multitask learning," in *Proc. 26th ICML*, 2009, pp. 1113–1120.

[5] T. N. Kipf and M. Welling, "Semi-supervised classification with graph convolutional networks," in *Proc. ICLR*, 2017.

[6] F. Wu, A. Souza, T. Zhang, C. Fifty, T. Yu, and K. Weinberger, "Simplifying graph convolutional networks," in *Proc. 36th ICML*, 2019, pp. 6861–6871.

[7] J. Platt, "Probabilistic outputs for support vector machines and comparisons to regularized likelihood methods," in *Advances in Large Margin Classifiers*, MIT Press, 1999, pp. 61–74.